FEU Institute of Technology
COLLEGE OF ENGINEERING • COLLEGE OF COMPUTER STUDIES

Computer System and Architecture

Machine Problem

MIDTERM

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Data Performed	Date Submitted

Paste the source code and explain the functionalities and take a screenshot of the final output.

Menu_loop - Handles the main program sequence. It prints the text menu options, requests your operation choice via sys_read (eax=3), and prompts for the two numbers.

cmp & je - the comparison of user input whether it is valid or invalid.

read_int - the user input reading

do_add - addiction function.

do_sub - Subtraction function

do_mul - multiplication function

do_div - Division function

invalid_choice - resets if invalid input or logical error like divisible by 0

print_int - output

ask_again - ask the user if they want to repeat

exit_program - termination

```
my_pc@DESKTOP-K8THJ9A:/mnt/c/Users/My PC$ ./calculator
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: d
Enter first number: 10
Enter second number: 10
The result is: 1

Do you want to try again (Y/N): y
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: d
Enter first number: 23
Enter second number: 0

Do you want to try again (Y/N): y
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: d
Enter first number: 0
Enter second number: 23
The result is: 0

Do you want to try again (Y/N):
```

```
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: m
Enter first number: 4
Enter second number: 4
The result is: 16

Do you want to try again (Y/N): y
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: m
Enter first number: 40
Enter second number: 3
The result is: 120

Do you want to try again (Y/N): y
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: m
Enter first number: 0
Enter second number: 4
The result is: 0

Do you want to try again (Y/N):
```

```
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: s
Enter first number: 3
Enter second number: 4
The result is: 4294967295

Do you want to try again (Y/N): y
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: s
Enter first number: 55
Enter second number: 4
The result is: 51

Do you want to try again (Y/N): y
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: s
Enter first number: 29
Enter second number: 29
The result is: 0

Do you want to try again (Y/N):
```

```
my_pc@DESKTOP-K8THJ9A:/mnt/c/Users/My PC$ nasm -f elf32 calculator.asm -o calculator.o
my_pc@DESKTOP-K8THJ9A:/mnt/c/Users/My PC$ ld -m elf_i386 calculator.o -o calculator
my_pc@DESKTOP-K8THJ9A:/mnt/c/Users/My PC$ ./calculator
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: a
Enter first number: 2
Enter second number: 4
The result is: 6

Do you want to try again (Y/N): y
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: a
Enter first number: 33
Enter second number: 44
The result is: 77

Do you want to try again (Y/N): y
---------------------------------
MENU
[A]dd
[S]ubtract
[M]ultiply
[D]ivide
---------------------------------
Enter your choice: a
Enter first number: 99
Enter second number: 2
The result is: 101

Do you want to try again (Y/N):
```